SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I **K.YAMUNA(192511013)** *(Name / Reg No) certify that this submission is my original work and that I have adheresive the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

K.YAMUNA
Signature of the student:

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN) **Team1** (4)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management **Team 3**
4. Machine Learning-Based Error Detection in Networks **Team 6 (3)**
5. Digital Twin Networks and the OSI Model **Team 2**
6. 6G Communication Architecture Compared with the OSI Model **Team 5, Team 9 (2)**
7. Edge Computing and Reliable Data Transmission **Team 10 (3)**
8. Cloud Networking Through the OSI Architecture **Team 4**
9. Satellite Internet Communication Using OSI Layers **Team 7 (3)**
10. Future of Reliable Networking Beyond the Traditional OSI Model **Team 8 (3)**

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation

Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

CORE TECHNICAL EXPLANATION:

Cloud Computing Through the OSI Architecture

- Cloud uses networking for communication.
- Data passes through all OSI layers.
- Internet transfers requests.
- Cloud processes the request.
- Response returns to the user.

INTRODUCTION:

What is Cloud Computing?

- Delivers computing services through the Internet.
- Provides storage, servers and software.
- Users can access services anytime.
- Easy to scale computing resources.
- Reduces infrastructure cost.

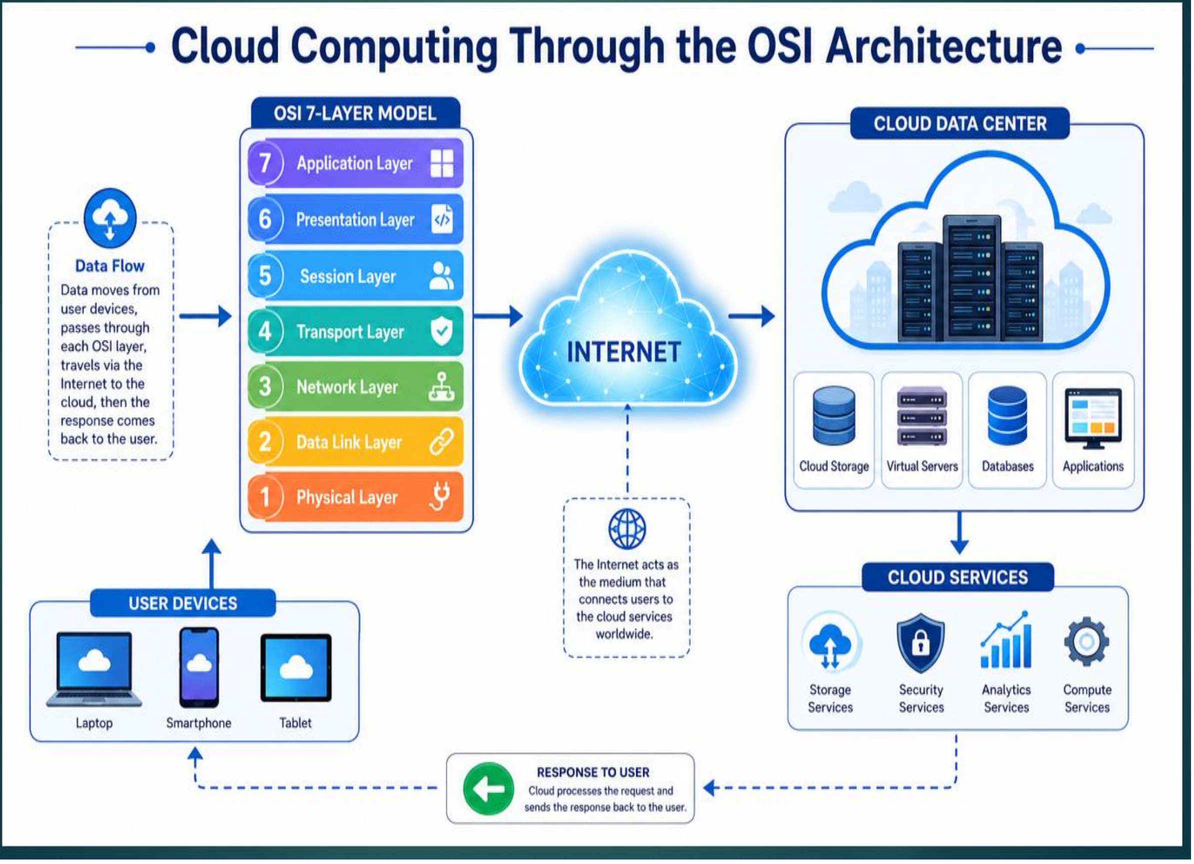
What is OSI Model?

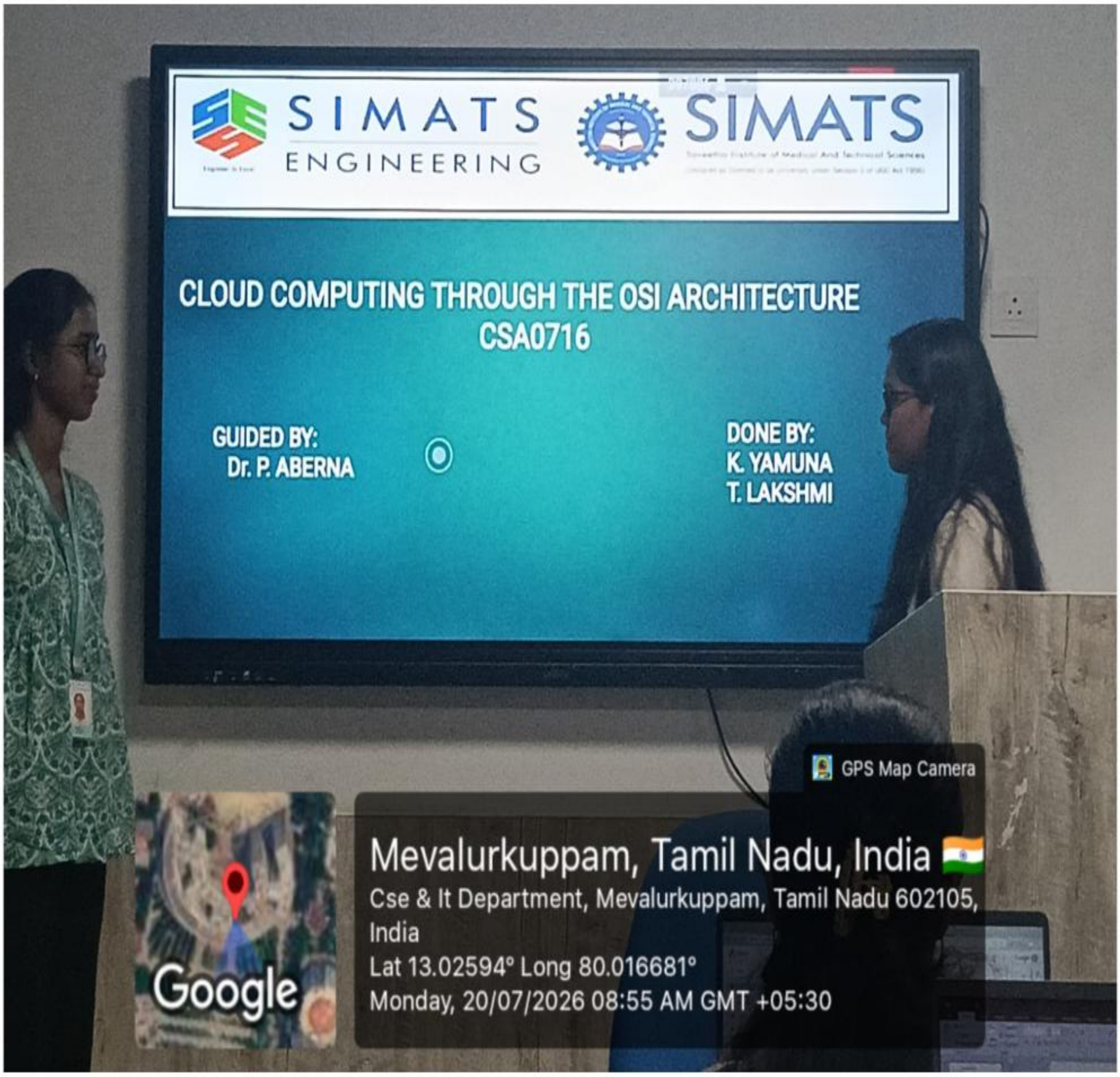
- Standard networking model.
- Contains seven layers.
- Defines communication between devices.
- Used in cloud networking.

ROLE OF OSI LAYERS:

OSI Layer	Role in Cloud Computing
Application	Gmail, Google Drive, Office 365
Presentation	Encryption and Compression
Session	User Login and Sessions
Transport	TCP and UDP Communication
Network	IP Routing
Data Link	Error Detection
Physical	Data Transmission

ARCHITECTURE DIAGRAM:




SIMATS
ENGINEERING
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CLOUD COMPUTING THROUGH THE OSI ARCHITECTURE
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